

Dollar-Cost Averaging Versus Lump-Sum-Investing: Evidence from France

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Abstract

1. Introduction

Dollar-Cost Averaging (DCA) is an investment strategy whereby an investor allocates capital in equal amounts at regular intervals over a predefined investment horizon. This progressive investment mechanism is generally motivated by its ability to reduce timing risk, particularly the risk of committing a large amount of capital at market peaks. By construction, DCA leads investors to purchase more shares when prices are low and fewer when prices are high, thereby smoothing the average purchase cost over time ([Dichtl and Drobetz, 2011](#)). From a theoretical perspective, DCA can also be interpreted as a risk management device, acting as a partial hedge against short-term market volatility by spreading entry points across time ([Milevsky and Posner, 2003](#)).

In contrast, Lump-Sum Investing (LSI) consists of investing the entire available capital at a single point in time, thus providing immediate and full exposure to market risk. In upward-trending markets, this strategy is often expected to outperform DCA, as capital is deployed earlier and benefits more rapidly from market appreciation. Consequently, LSI may appear more attractive to investors with strong bullish expectations. Conversely, DCA tends to be preferred in environments characterized by heightened uncertainty or anticipated downturns ([Grable and Chatterjee, 2015](#)). However, such preferences implicitly rely on the ability to time market conditions, which remains notoriously difficult in practice.

This tension naturally raises a fundamental question for investors: which strategy should be preferred when entering financial markets under uncertainty? More specifically, beyond conditional preferences based on market expectations, it becomes essential to assess which approach delivers superior performance on average, and how sensitive this performance is to the timing of market entry.

The objective of this study is therefore to provide a comparative analysis of DCA and LSI strategies. In particular, we aim to contribute to the existing literature by offering new empirical evidence based on the French market. Additionally, this work seeks to enrich the debate by linking investment outcomes to behavioral considerations, thereby providing a more comprehensive understanding of how investors navigate the trade-off between risk and return when choosing between these two widely used investment strategies.

2. Methodology

2.1 Investment Framework and Assumptions

This study compares the performance of Dollar-Cost Averaging (DCA) and Lump-Sum Investing (LSI) on the French equity market. Both strategies are evaluated under a common set of assumptions, consistent with the existing literature (Leggio and Lien, 2001; Eriksson and Fransson, 2021).

An investor is assumed to have a fixed initial capital W_0 available at time $t = 0$. This capital represents a lump sum that could, in principle, be deployed immediately or phased into the market over a predefined horizon. The investor selects a single stock listed on Euronext Paris and commits to one of the two strategies without any subsequent tactical reallocation. No transaction costs are considered, and all dividends are assumed to be reinvested continuously, as reflected in the use of total return price series. Returns are expressed in nominal terms, with no adjustment for inflation. The uninvested cash under the DCA strategy is assumed to earn zero interest, consistent with the approach adopted by Eriksson and Fransson (2021) and representative of the current low-yield savings environment faced by retail investors in France. This assumption is conservative with respect to DCA, as it does not credit the cash reserve with any return during the phasing-in period, thereby introducing a mild bias in favour of LSI. This limitation is acknowledged and discussed in Section limitations.

2.2 Strategy Definitions

2.2.1 Lump-Sum Investing (LSI)

Under the LSI strategy, the entire capital W_0 is invested at the beginning of the investment window, at time $t = 0$, at the observed market price P_0 . The investor acquires

$$N^{\text{LSI}} = \frac{W_0}{P_0}$$

shares and holds them until the end of the horizon at time $t = T$. Since dividends are fully reinvested, the total return price series captures all forms of value creation over the holding period. The Holding Period Return (HPR) is defined as:

$$\text{HPR}^{\text{LSI}} = \frac{P_T^{\text{TR}} - P_0^{\text{TR}}}{P_0^{\text{TR}}}$$

where P_t^{TR} denotes the total return adjusted price at time t . By construction, LSI provides immediate and full market exposure from $t = 0$, and the investor therefore benefits — or suffers — from any price movement over the entire horizon.

2.2.2 Dollar-Cost Averaging (DCA)

Capital W_0 is split into N equal tranches $w = W_0/N$, deployed at dates $\{t_1, \dots, t_N\}$ with $t_1 = 0, t_N = T$. At each t_k , the tranche buys

$$n_k = \frac{W_0/N}{P_{t_k}^{\text{TR}}}$$

shares at the total-return-adjusted price $P_{t_k}^{\text{TR}}$, which implicitly compounds all dividends and eliminates any separate reinvestment step. The fixed outlay w means share quantities vary inversely with price.

After the k -th deployment, cumulative holdings and mark-to-market value are

$$S_k = \frac{W_0}{N} \sum_{j=1}^k \frac{1}{P_{t_j}^{\text{TR}}}, \quad V_{t_k}^{\text{DCA}} = S_k \cdot P_{t_k}^{\text{TR}}.$$

Full investment is reached only at $t_N = T$, yielding terminal shares S_N and value $V_T^{\text{DCA}} = S_N \cdot P_T^{\text{TR}}$. The Holding Period Return is

$$\text{HPR}^{\text{DCA}} = \frac{V_T^{\text{DCA}} - W_0}{W_0}.$$

2.3 Rolling Window Simulation

Rather than evaluating the strategies over a single fixed period, this study employs a rolling window approach, consistent with the methodology of Leggio and Lien (2001) and Eriksson and Fransson (2021). For a given investment horizon H (expressed in months), a window of length H is rolled forward one month at a time over the full sample period. This generates a distribution of HPR outcomes for both strategies across all possible entry points, thereby mitigating the sensitivity of results to any particular starting date and accounting for the full range of market conditions — including expansions, contractions, and crisis episodes — present in the data.

Formally, let $\mathcal{T} = \{s_1, s_2, \dots, s_M\}$ denote the set of monthly window start dates, where M is the total number of complete windows of length H contained in the sample. For each start date s_i , we compute $\text{HPR}_i^{\text{LSI}}$ and $\text{HPR}_i^{\text{DCA}}$ as defined above, yielding two empirical distributions of size M that form the basis for all subsequent statistical comparisons.

Four investment horizons are considered: $H \in \{12, 24, 36, 48, 64, 72\}$ months, following the parametrisation of Eriksson and Fransson (2021). The number of rolling windows decreases as H increases, since longer horizons consume more of the available sample.

2.4 Performance Measures

The relative performance of DCA and LSI is assessed through four complementary metrics, each capturing a distinct dimension of investment quality.

2.4.1 Holding Period Return (HPR)

The HPR measures the raw total return generated over the investment horizon. Although it does not account for risk, it provides a direct measure of wealth creation and serves as the primary basis for comparing the two strategies. The mean HPR across all rolling windows is the central quantity of interest:

$$\overline{\text{HPR}} = \frac{1}{M} \sum_{i=1}^M \text{HPR}_i$$

2.4.2 Sharpe Ratio

The Sharpe ratio measures risk-adjusted performance from the perspective of a mean-variance investor (Markowitz, 1952; Sharpe, 1966). It is computed over the cross-section of rolling window returns as:

$$SR = \frac{\overline{HPR} - r_f^{(H)}}{\sigma(HPR)}$$

where $r_f^{(H)} = (1 + r_f)^{H/12} - 1$ is the risk-free rate compounded over the horizon H , and $\sigma(HPR)$ is the standard deviation of HPR across all rolling windows. A higher Sharpe ratio indicates greater excess return per unit of total risk.

2.4.3 Sortino Ratio

The Sortino ratio refines the Sharpe ratio by restricting the risk measure to downside deviations only (Sortino and Price, 1994). This is particularly relevant in the context of prospect theory, where losses are perceived as more painful than equivalent gains (Kahneman and Tversky, 1979). It is defined as:

$$SoR = \frac{\overline{HPR} - r_f^{(H)}}{\sigma_d(HPR)}$$

where the downside deviation is:

$$\sigma_d = \sqrt{\frac{1}{M} \sum_{i=1}^M \left[\min\left(0, HPR_i - r_f^{(H)}\right) \right]^2}$$

This measure penalises only returns that fall below the risk-free benchmark, leaving upside volatility unrewarded.

2.5 Statistical Testing

For each asset, horizon, and DCA frequency, the null hypothesis that the mean HPR of LSI equals that of DCA is tested using a paired t -test:

$$H_0 : \mu_{LSI} - \mu_{DCA} = 0 \quad \text{vs.} \quad H_1 : \mu_{LSI} - \mu_{DCA} \neq 0$$

The paired structure exploits the fact that each observation (HPR_i^{LSI} , HPR_i^{DCA}) is computed over the same window $[s_i, s_i + H]$, thereby controlling for window-specific market conditions and yielding a more powerful test than an independent-samples alternative. Statistical significance is reported at the 1% (***), 5% (**), and 10% (*) levels.

In addition to the t -test, the LSI win rate is reported for each configuration:

$$\text{Win Rate}_{LSI} = \frac{1}{M} \sum_{i=1}^M \mathbf{1}[HPR_i^{LSI} > HPR_i^{DCA}]$$

This non-parametric measure directly quantifies the fraction of historical entry points for which LSI would have been the preferable strategy, offering an intuitive complement to the mean-based tests.

References

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